1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlights that local immune architecture (Immunoscore, leukocyte fraction) and MMR status (MSH6) jointly influence NMIBC prognosis, but immune-rich tumors do not always exhibit favorable outcomes—why does organized local immunity (TLS) sometimes fail to prevent progression in NMIBC? 2) Tier 1 paths used: I built on the Immunoscore/leukocyte-fraction axis (immune contexture) together with MSH6 (MMR/genomic instability) because these provide a direct mechanistic link between tumor immunogenicity, local immune organization, and clinical outcome. 3) Tier 2 evidence that supports the claim: systemic/metabolic and noninvasive readouts (HDL-C, urine DNA methylation, stool SDC2/TFPI2) and AI-derived integrative signatures plausibly modulate or report on immune competence and tumor visibility, and thus can act as modifiers or measurement tools in follow-up studies. 4) Why PFI: For NMIBC the key clinical problem is recurrence/progression rather than immediate OS; progression-free interval is the most relevant, sensitive endpoint to capture the clinical consequence of failed local immune control. 5) What I deliberately excluded and why: I excluded exploratory/weakly connected paths (CSF AD biomarkers, EV-MDK comparisons, PNI as primary anchors) and did not include multiple immune-checkpoint or other gene-level modifiers in the core hypothesis to keep the mechanistic claim sharply focused on how MSH6-related genomic context modulates TLS effectiveness via leukocyte functional state?

Working Hypothesis

TLS-like lymphoid aggregates are associated with longer progression-free interval (PFI) in NMIBC only within a functional immune context defined by intact MSH6 status and a sufficiently high intratumoral leukocyte fraction. Specifically, MSH6-proficient tumors with TLS + high leukocyte fraction will show the longest PFI, whereas MSH6-deficient tumors or tumors with low leukocyte fraction will not derive the same TLS-associated benefit. TLS presence in the setting of MSH6 deficiency may even coincide with shorter PFI due to elevated neoantigen burden driving T-cell exhaustion within TLS. Therefore, there is a significant three-way interaction (TLS status × MSH6 status × leukocyte fraction) that predicts PFI in NMIBC, reconciling variable TLS prognostic value observed across tumors and enabling a stratified prognostic model using only existing TLS imaging features and Tier 1 anchors.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	MSH6, Leukocyte fraction
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6) jointly influence NMIBC prognosis, but immune-rich tumors do not always exhibit favorable outcomes—why does organized local immunity (TLS) sometimes fail to prevent progression in NMIBC? 2) Tier 1 paths used: I built on the Immunoscore/leukocyte-fraction axis (immune contexture) together with MSH6 (MMR/genomic instability) because these provide a direct mechanistic link between tumor immunogenicity, local immune organization, and clinical outcome.

Leukocyte fraction: leukocyte fraction) and MMR status (MSH6) jointly influence NMIBC prognosis, but immune-rich tumors do not always exhibit favorable outcomes—why does organized local immunity (TLS) sometimes fail to prevent progression in NMIBC? 2) Tier 1 paths used: I built on the Immunoscore/leukocyte-fraction axis (immune contexture) together with MSH6 (MMR/genomic instability) because these provide a direct mechanistic link between tumor immunogenicity, local immune organization, and clinical outcome.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	Immunoscore®	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	Urine DNA methylation biomarkers	Supporting modifier from knowledge graph
Secondary	AI-based biomarker signatures	Supporting modifier from knowledge graph
Secondary	SDC2	Supporting modifier from knowledge graph
Secondary	TFPI2	Supporting modifier from knowledge graph
Secondary	Prognostic Nutritional Index (PNI)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: assemble a cohort with histology-scored TLS, MSH6 IHC/sequence data, and leukocyte fraction (via pathology estimates or transcriptomics).

Prediction 2: Define TLS status (present/absent) and categorize MSH6 as proficient/deficient.

Prediction 3: Model PFI with Cox regression including TLS, MSH6 status, leukocyte fraction (continuous or categorized), and all two-way and three-way interaction terms.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- Immunoscore®: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- HDL-C: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.